AYOMIDE ABILEWA

Lekki, Lagos, Nigeria · abilewaayomide@gmail.com · 09066021641
linkedin.com/in/ayomide-abilewa-443aa9313 · github.com/ayomide-abilewa · ayomide-abilewa.github.io

Profile

Builds embedded and computer-vision systems, with a habit of designing for the conditions that break them: poor lighting, no network, unreliable input. Interned on the electrical and instrumentation team at Chevron Nigeria and worked with the control loops and signal standards behind the same measurement problems at industrial scale.

Technical Skills

Programming: Python, C/C++, MATLAB, JavaScript, Verilog/HDL, Bash

Embedded and Hardware: Arduino, ESP32, STM32, Raspberry Pi, FPGA, I2C / SPI / UART, Sensor integration

AI and Machine Learning: YOLOv8/v11, OpenCV, TensorFlow (fundamentals), PyTorch (fundamentals), Weighted Boxes Fusion

Instrumentation and Control: P&ID interpretation, PID control loops, Pneumatic systems, I/P converters, 4–20 mA and 3–15 PSI standards

Electrical Systems: SLD interpretation, Circuit breaker sizing, ESP systems, Power transmission, Arc flash safety, Hazardous area classification

Tools: VS Code, Git/GitHub, MATLAB/Simulink, KiCad, Flask, React/TypeScript, PostgreSQL, Blynk IoT

Engineering Experience

Facilities Engineering Management Intern (SIWES) Apr–Sep 2026

Chevron Nigeria Limited, Lagos, Nigeria

- Rotated through the Electrical and Instrumentation (E&I) team on field instrumentation and electrical systems.
- Read P&IDs before fieldwork; traced control loop configurations, I/P converters and 3–15 PSI pneumatic signalling.
- On the electrical side: single line diagrams, ESP systems, power transmission, circuit breaker specification.

Research and Development Intern (SIWES) 2024

ACE Quanser Robotics Lab, Obafemi Awolowo University, Ile-Ife, Nigeria

- Operated three Quanser platforms in MATLAB and Simulink: the QDrone quadrotor, the QCar ground vehicle, the QArm 6-DOF manipulator.
- PID control and sensor feedback tuned on live hardware, not in simulation.

Embedded Systems and EEE Tutor 2023–Present

Astar Tutorials, Ile-Ife, Nigeria

- Teach circuit analysis, digital logic and microcontroller programming to engineering undergraduates.

Selected Projects

aniwe Aug 2026

Python, Streamlit, OpenRouter, Vision-language models

- Shipped a convention costume-rating booth in Python and Streamlit: ten hosts, per-session mood and quirk variation, vision-language scoring over seven criteria.

- Text-to-speech falls through three tiers (ElevenLabs, offline Piper, browser speech) and speech-to-text runs locally, so a blocked network costs quality and nothing else.
- 214 offline tests, 1.5 seconds, no network or camera required — truncated-JSON repair, persona rotation, rank boundaries, the whole fallback chain.

github.com/ayomide-abilewa/aniwe

AI-Powered Smart Attendance System Feb–Mar 2026

Python, OpenCV, YuNet, SFace

- Owned the vision pipeline and the Flask REST API: YuNet detection, SFace recognition, cosine-similarity matching.
- Built the React and TypeScript frontend over a PostgreSQL schema for students, courses, enrolment, sessions and attendance.
- Defended it at the final-year project review; invited afterwards to build a production version.

github.com/ayomide-abilewa/Smart-Attendance-System

Live Video Pipe Anomaly Detection 2026–Present

Python, YOLOv8, YOLOv11, Weighted Boxes Fusion

- Building a two-stage cascaded detector for real-time pipe anomaly identification in industrial inspection.
- Two YOLO models run in parallel, one on RGB and one on edge-enhanced input, merged with Weighted Boxes Fusion to hold precision across lighting and surface conditions.

Multispectral Acquisition Device for Hydroponics 2025

ESP32, GY-2561, GY-31, I2C

- Built an ESP32 acquisition node measuring light intensity and colour spectrum with the GY-2561 and GY-31, streaming live to a Blynk dashboard.
- Two I2C sensors on one bus, with arbitration and calibration handled so the readings mean something in absolute terms.

Education

B.Sc. Electronic and Electrical Engineering 2021–2027 (expected)

Obafemi Awolowo University, Ile-Ife, Osun State, Nigeria

Training and Certifications

Operational Excellence Training, Chevron 2026

Cyber Security and Data Privacy Awareness, Chevron 2026

FPGA Design and HDL/Verilog Workshop (SWEP), Faculty of Engineering, Obafemi Awolowo University 2024

Robotics Lab Orientation and Operations, ACE Quanser Lab, Obafemi Awolowo University 2024

Leadership and Teaching

SPAW Project Lead and Syllabus Team Lead 2024–Present

IEEE OAU Student Branch

- Lead SPAW 3.0, a six-session embedded systems and entrepreneurship curriculum for secondary school students.